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Introduction:

This project builds upon the existing Genetic Algorithm/Creatures case study, which originally explored the evolution of simulated modular creatures within a virtual environment. The foundational system employed genetic encoding to procedurally generate articulated bodies using a simplified genome, allowing evolutionary processes such as crossover, mutation, and selection to optimize physical forms for movement-based tasks. In the original setup, the fitness of each creature was evaluated primarily by how far it could travel across a flat plane.

The goal of this extended coursework is to adapt the existing evolutionary framework so that the creatures evolve to complete a more complex task: climbing a mountain. This task introduces new environmental constraints and fitness considerations, such as vertical mobility and terrain navigation, which were absent in the original flat-world environment. In doing so, this work aims to assess the capacity of evolutionary strategies to produce morphological and behavioural adaptations in response to more challenging terrain.

By integrating the genetic algorithm with a new mountainous terrain, this project offers insight into how evolutionary computation can solve more sophisticated physical problems. The outcomes may also serve as a simplified model for understanding evolutionary dynamics in variable and non-uniform environments.

Environment Setup:

To facilitate the new climbing task, a custom three-dimensional terrain was developed and integrated into the simulation environment. The simulation runs on the PyBullet physics engine, chosen for its real-time capabilities, support for URDF-based robotic structures, and extensive integration with Python. The environment consists of a bounded arena with four vertical walls and a flat base, onto which a mountain structure is placed.

The mountainous terrain is modelled as a Gaussian-shaped pyramid, procedurally generated using Perlin noise and Gaussian functions. This terrain was exported into OBJ format and subsequently wrapped in a URDF file to enable loading into PyBullet. The elevation of the terrain gradually increases toward the centre, providing a continuous incline that mimics real-world climbing conditions. The terrain is fixed in place and does not respond to physical contact, ensuring that any climbing movement arises purely from the creature's own locomotion capabilities.

Creatures are initially spawned at a lower elevation near the base of the mountain. Their progress is evaluated based on two main metrics: the maximum vertical height reached and the proximity to the central peak. The simulation supports both graphical (GUI) and headless (DIRECT) modes, with the latter used for large-scale evolutionary experiments to increase computational efficiency.

The environment setup was implemented in such a way that it supports repeated resets between creature evaluations, ensuring that each individual is tested under consistent conditions. This

enables fair fitness comparisons across generations and parameter configurations during the evolutionary process.

Genetic Algorithm Implementation:

The genetic algorithm (GA) used in this project is adapted from the original Creatures case study and implemented within `mov-envt.py`. Its objective is to evolve modular creatures capable of climbing a procedurally generated mountain.

Each creature is encoded as a genome comprising multiple genes, with each gene controlling structural and motor parameters such as link length, joint orientation, and control signals. These genomes are transformed into URDF-based articulated bodies for simulation in PyBullet. The GA employs three mutation operators: point mutation (adjusting gene values), grow mutation (adding genes), and shrink mutation (removing genes). Crossover is applied by combining segments of two parent genomes. Selection is based on fitness-proportionate (roulette wheel) sampling.

Fitness is calculated from two criteria: the maximum height a creature reaches and its proximity to the mountain centre. After evaluation, a new generation is formed through reproduction and mutation. Elitism is used to carry the best-performing individual into the next generation.

This generational loop continues for a set number of iterations, with all simulation and evaluation steps automated and run in headless mode for efficiency.

Basic Experiment:

The basic experiment investigates how different evolutionary parameters influence the ability of simulated creatures to climb a mountain. The focus is on evaluating the best fitness score achieved under varying configurations of four key factors: population size, genome size, mutation rate, and number of generations.

The default (baseline) configuration is:

- **Generations:** 100
- **Population Size:** 20
- **Genome Size:** 10 genes
- **Mutation Rate:** 0.01

Each parameter was varied independently while keeping the others fixed to the baseline. The values tested were as follows:

- **Population Size:** 20, 30, 40

- **Genome Size:** 10, 15, 20
- **Mutation Rate:** 0.01, 0.5, 0.8
- **Number of Generations:** 100, 200, 300

For each setting, the evolutionary simulation was run using **mov-envt.py**, and the best fitness score from the final generation was recorded. Fitness was calculated as a combination of maximum height reached and proximity to the mountain's centre, with higher scores indicating better climbing performance.

The results are used to assess which factors contribute most significantly to evolutionary success, and to identify optimal configurations for evolving effective climbers.

Fitness Score Calculation

Each creature's performance was evaluated based on its ability to climb the mountain and move toward its centre. The fitness score was computed as:

- **Height Fitness** = Maximum height (Z-axis) reached
- **Proximity Fitness** = $\max(0, 15 - \text{distance to mountain centre})$
- **Total Fitness** = Height Fitness + $0.5 \times$ Proximity Fitness

This composite score rewards both vertical climbing and accurate navigation.

Fitness Score Ranges

- **Theoretical Minimum:** 0
 - Creature fails to generate or falls immediately
- **Practical Worst (~5.0–8.0):**
 - Minimal or no movement after spawning
- **Basic Locomotion (8–10):**
 - Moves toward the mountain base but does not climb
- **Good Climber (10–12):**
 - Successfully climbs partway up the mountain
- **Elite Climber (12–12.5):**
 - Reaches the mountain peak and remains near the centre
- **Theoretical Maximum:** 12.5
 - Height = 5.0 (peak), Distance = 0.0 (center)

This scoring framework provides a consistent way to evaluate and compare the effects of different evolutionary parameters on climbing performance.

Advanced Experiment:

The advanced experiment investigates the role of new motor types in evolving effective climbing behaviour. Two new motor waveforms — triangle and random — were added to the simulation

alongside the existing pulse and sine motors. These were introduced to explore whether alternative control signals could produce better climbing adaptations or unlock previously unseen behavioural strategies.

The experimental setup maintains the base configuration for consistency with the basic experiment:

- **Generations: 100**
- **Population Size: 20**
- **Genome Size: 10**
- **Mutation Rate: 0.01**

Motor Type Variants

Each creature's joints are controlled by motors whose behaviour is dictated by a waveform gene. The available motor types now include:

- **Pulse (binary switching)**
- **Sine (smooth oscillation)**
- **Triangle (linear rising and falling ramp)**
- **Random (chaotic, non-repeating values)**

To isolate the effects of the new motor types, the experiment was conducted in three distinct phases:

Triangle-Only Test

All motors were forced to use triangle waveforms. This created consistent, smooth motion with gradual acceleration and deceleration, potentially beneficial for climbing by offering steady force application and better traction.

Random-Only Test

All motors were forced to use random waveforms. These motors produced unpredictable movement patterns, rapidly switching direction and speed. While generally inefficient, this configuration tested whether high exploratory behaviour might occasionally yield unexpected climbing solutions.

Full Motor Diversity

The final phase allowed all four motor types to be selected probabilistically based on the control_waveform gene. This enabled the evolutionary process to select from a diverse set of movement strategies and potentially evolve hybrid behaviours that combine precision and exploration.

Results:

Basic Experiments results

Testing Population Size									
Run ID	Population Size	Initial Genome Size	Mutation Rate	No of Generation	Initial Average Fitness	Initial Best Fitness	Final Average Fitness	Final Best Fitness	
	1	20	10	0.01	100	5.489	6.248	6.362	7.592
	2	30	10	0.01	100	5.618	15.893	7.073	29.639
	3	40	10	0.01	100	5.578	7.149	5.734	7.862
Testing Initial Genome Size									
Run ID	Population Size	Initial Genome Size	Mutation Rate	No of Generation	Initial Average Fitness	Initial Best Fitness	Final Average Fitness	Final Best Fitness	
	1	20	10	0.01	100	5.489	6.248	6.362	7.592
	2	20	15	0.01	100	2.898	6.264	6.229	7.763
	3	20	20	0.01	100	1.424	5.932	5.771	7.211
Testing Mutation Rate									
Run ID	Population Size	Initial Genome Size	Mutation Rate	No of Generation	Initial Average Fitness	Initial Best Fitness	Final Average Fitness	Final Best Fitness	
	1	20	10	0.01	100	5.489	6.248	6.362	7.592
	2	20	10	0.5	100	5.282	6.987	5.821	7.895
	3	20	10	0.8	100	5.967	6.939	7.052	31.336
Testing No of Generation									
Run ID	Population Size	Initial Genome Size	Mutation Rate	No of Generation	Initial Average Fitness	Initial Best Fitness	Final Average Fitness	Final Best Fitness	
	1	20	10	0.01	100	5.489	6.248	6.362	7.592
	2	20	10	0.01	200	5.853	6.18	5.75	7.676
	3	20	10	0.01	300	5.489	6.498	8.727	31.892

Population Size

Increasing the population size from 20 to 30 led to a substantial improvement in final best fitness (7.592 → 29.639). However, increasing it further to 40 showed a drop in performance (final best fitness: 7.862), possibly due to greater genetic diversity slowing convergence.

Hypothetical insight: A moderate population size (30) may provide a balance between exploration and exploitation, promoting faster discovery of beneficial traits.

Genome Size

Larger genome sizes (15 and 20 genes) resulted in lower starting fitness and only slight improvements by the end of evolution. The best result was still achieved with the baseline 10-gene configuration.

Hypothetical insight: Smaller genomes might be easier to evolve initially, while larger genomes could introduce unnecessary complexity that slows adaptation in early generations.

Mutation Rate

Higher mutation rates yielded stronger improvements. While 0.5 slightly outperformed the baseline, the 0.8 mutation rate produced a final best fitness of **31.336**, indicating more diverse structural adaptations.

Hypothetical insight: A high mutation rate may help escape local optima and accelerate discovery of climbing behaviors, though it could also introduce instability in longer runs.

Number of Generations

Extending the number of generations significantly improved performance. The final best fitness rose from 7.592 (100 generations) to **31.892** (300 generations), showing continued gains with time.

Hypothetical insight: More generations give the evolutionary process time to refine traits and accumulate beneficial mutations, especially in tasks requiring complex coordination.

Advanced Experiment results

Motor Types	Initial Average Fit	Initial Best Fitness	Final Average Fitness	Final Best Fitness
Triangle Motor Only	5.25	6.444	5.832	7.599
Random Motor Only	5.179	6.427	5.839	7.639
Normal Motor Selection Between the 4 types	6.021	9.718	5.909	10.905

Motor Type

Three configurations were tested: triangle motor only, random motor only, and normal selection between all four motor types (pulse, sine, triangle, and random). Both triangle and random motors produced only modest fitness gains across 100 generations. Triangle motors resulted in a final best fitness of **7.599**, while random motors achieved a slightly higher value of **7.639**. In contrast, allowing normal selection among all four motor types led to a significantly higher final best fitness of **10.905**.

Hypothetical insight: Constraining the creatures to a single motor type may limit the behavioural repertoire available for climbing. Triangle motors offer predictable control but may lack adaptability, while random motors promote exploration but suffer from instability. In contrast, allowing evolution to choose among all available waveforms enables more flexible and specialised motor strategies, potentially leading to more effective locomotion and coordination.

Images:

Starting Process

```
pybullet build time: May 27 2025 11:28:06
Starting evolution with your specified parameters:
  Generations: 100
  Population size: 20
  Initial genome size: 10 genes
  Mutation rate: 0.8
=== Mountain Climbing Evolution Parameters ===
Population size: 20
Initial genome size: 10 genes
Mutation rate: 0.8
Number of generations: 100
=====

Initializing population...
```

Process of evolving the creatures

```
=== Generation 19/100 ===
Evaluated creature 1/20
Evaluated creature 6/20
Evaluated creature 11/20
Evaluated creature 16/20
Warning: Lost creature 0 at step 0: GetBasePositionAndOrientation failed.
Best fitness: 18.094
Average fitness: 6.355
Max height reached: 13.010
Min distance to center: 4.832
Mean links: 38.0
Elite saved to mountain_climber_gen_19.csv
```

Result of completing the evolution of the creatures

```
=====
EVOLUTION COMPLETE - FINAL RESULTS
=====
Total generations run: 100
Final best fitness: 31.336
Final average fitness: 7.052
Initial best fitness: 6.939
Initial average fitness: 5.967
Improvement in best fitness: 24.397
Improvement in average fitness: 1.086

Fitness progression (every 50 generations):
Generation | Best Fitness | Avg Fitness
-----
1 | 6.939 | 5.967
51 | 31.206 | 7.113
```

Images of Creatures



